

3altayer.app - Project Specification & Architecture

Hybrid Ride-Hailing Platform (Instant & Bidding) | Currency: EGP (Egyptian Pounds)

1. Executive Overview & Hybrid Mechanism

- **Core Vision:** Hybrid ride-hailing combining Instant (Uber-like) and Bidding (inDrive-like) modes.
- **Instant Ride Mode:** Fixed calculated upfront fare, auto-dispatch to nearest driver within 15s timeout. (15% commission)
- **Bidding Ride Mode:** Recommended benchmark fare, driver counter-offers in real-time, rider picks preferred driver. (10% commission)
- **Primary Target Market:** Egypt (Greater Cairo & Alexandria) | Primary Language: Arabic (RTL), Secondary: English.

2. Fleet Categories & Pricing Structure in Egypt (EGP)

- **Economy (Sedan):** Base: 18.00 EGP | Per KM: 5.00 EGP | Minimum: 25.00 EGP
- **Comfort (Air-Conditioned):** Base: 28.00 EGP | Per KM: 7.00 EGP | Minimum: 40.00 EGP
- **Scooter / Motorcycle:** Base: 12.00 EGP | Per KM: 3.50 EGP | Minimum: 18.00 EGP
- **Toktok (Local Short-Trips):** Base: 10.00 EGP | Per KM: 3.00 EGP | Minimum: 15.00 EGP

3. Technical Infrastructure & Architecture

- **Backend API:** Node.js (NestJS + TypeScript) - Modular enterprise architecture with REST & Swagger.
- **Real-time & Geospatial:** Redis Geo for ultra-low latency driver coordinates & WebSocket (Socket.io) broadcast.
- **Database & ORM:** PostgreSQL 16 + PostGIS spatial extensions managed by Prisma ORM.
- **Admin Dashboard:** Next.js 14 + Tailwind CSS with Live Radar Visualizer & bilingual Arabic/English support.
- **Mobile Ecosystem:** Flutter (Dart) shared monorepo for Rider and Driver apps on iOS & Android.

4. Safety Protocols & Financial Ledger (Wallets)

- **Ride Security:** 4-digit OTP code verified before trip starts, SOS emergency button, and live trip sharing.
- **Cash Settlements:** Automatic platform commission debit from driver in-app wallet upon trip completion.
- **Driver Debt Limit:** Allowed negative debt limit up to -150.00 EGP before requiring a top-up recharge.
- **Digital Payments:** Instant net earnings credit for electronic card and mobile wallet transactions.

5. Milestones & Delivery Status

- **Phase 0 & 1 [Completed]:** Docker, PostgreSQL, Redis, Auth OTP, JWT, and Driver profiles established.
- **Phase 2 & 3 [Completed]:** Geospatial Redis Geo engine, Instant auto-dispatch, Bidding broadcast & OTP lifecycle.
- **Phase 4 & 5 [Completed]:** Wallet ledger in EGP, Admin Operations radar, and bilingual support (Arabic primary).
- **Phase 6 [Next Steps]:** Flutter mobile UI screen completion and staging soft launch.